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SURVEY

Transforming Education With Generative AI: A Comprehensive Review of Advancements, Challenges, and Future Opportunities

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ABSTRACT The swift progress of technology and the rise of generative artificial intelligence (generative AI) have significantly transformed educational paradigms, necessitating a thorough examination of their impacts and applications in learning environments. This systematic literature review aims to explore the evolving landscape of generative AI in education by analyzing a comprehensive selection of 55 studies indexed in Web of Science (WoS), selected for its high-quality peer-reviewed content and multidisciplinary coverage, that investigate the effectiveness of generative AI tools in enhancing educational outcomes. This review highlights their role in personalized learning experiences, adaptive assessments, and automated feedback systems. Our analysis highlights the potential of generative AI to foster engagement through interactive and immersive learning environments while also addressing the challenges of traditional educational models that often neglect individual learning styles and needs. Key themes emerged regarding the incorporation of AI technologies, including natural language processing and chatbot applications, which have proven beneficial in improving academic skills and optimizing administrative processes. This review also identifies pressing challenges requiring further exploration, including the ethical concerns surrounding data privacy and biases in algorithmic decision processes. These findings offer meaningful insights for educators and researchers, guiding future work toward leveraging generative AI to create more inclusive and impactful learning experiences.

INDEX TERMS Generative artificial intelligence, artificial intelligence in education, personalized learning, adaptive assessment, automated feedback, natural language processing, chatbot applications, ethical implications, data privacy, algorithmic bias.

I. INTRODUCTION

The roots of modern e-learning can be traced back to pioneering experiments in computer-based training (CBT) during the 1960s and 1970s [1]. However, it was the rise of the internet that truly accelerated its evolution, catalyzing a significant shift in educational paradigms [2].

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Concepts such as asynchronous and synchronous learning emerged as foundational elements, offering the flexibility of real-time interactions or self-paced engagement [3], [4]. The development of blended learning further enriched the educational experience by merging online tools with face-to-face learning instruction [5]. This evolution includes a diverse array of online educational formats, encompassing web-based classes and digital learning environments, and immersive and dynamic educational resources [6].

Additionally, the emergence of Massive Open Online Courses (MOOCs) have broadened access to top-tier educational resources, reaching learners from around the globe [7], [8], [9]. Learning Management Systems (LMS) quickly became essential for organizing and distributing digital learning materials, enabling instructors to track progress as well as evaluate performance [10]. Moreover, the advent of mobile learning revolutionized accessibility, empowering learners to engage with content via smartphones and tablets [11]. These developments, along with the rise of artificial intelligence (AI), gamification, and interactive multimedia, have reshaped the educational landscape [12]. These trends reflect the evolving landscape of e-learning, shaped by technological innovation and the global movement towards democratizing education.

Recent reviews also highlight how generative AI offers powerful means of personalizing instruction and automating feedback while raising new challenges around bias and data privacy [13].

Traditional educational models have long been critiqued for relying on standardized curricula and teaching methods, which often overlook students' varied learning preferences, capabilities, and personal interests. This approach has been associated with disengagement and a lack of opportunities for personal growth [14], [15], [16]. The absence of personalized learning experiences further hinders students from achieving their full potential [16], [17]. Additionally, traditional education has been criticized for its emphasis on rote memorization at the expense of developing critical thinking, problem-solving, and creativity [18]. The teacher-centered nature of this model often diminishes student autonomy, stifling the cultivation of essential critical thinking skills [19].

Generative AI, a branch of artificial intelligence aimed at producing original content such as text, images, or audio by leveraging recognized patterns derived from data [20], [21], indicates a promising solution to many of the difficulties related to traditional education by offering innovative, adaptive, and interactive learning environments. One major advantage of generative AI is its capacity to customize learning experiences by adapting content and responses to each learner's requirements based on their learning preferences and progress [22], [23], [24]. Furthermore, AI-powered interactive tools and virtual assistants foster engagement, promoting immersive learning experiences that encourage active student participation [22], [24]. The flexibility of generative AI extends to assessments, automated grading, and feedback mechanisms offering timely as well as personalized evaluations [25]. Moreover, AI-driven tools support teachers by automating administrative tasks, analyzing student data, and providing personalized assistance, allowing educators to devote more time to meeting the specific needs of their students [25].

The contribution of generative AI within the educational field is steadily expanding, improving educational experiences, streamlining school operations, and offering tailored

assistance. Research has demonstrated the effectiveness of artificial intelligence tools, including NLP and conversational agents, in addressing various educational needs, including improving academic writing [26], fostering computational thinking [27], and streamlining administrative tasks [28], [29]. Automated feedback systems, for example, have proven effective in enhancing language learning by improving writing skills, while tools like ChatGPT [30] have demonstrated their ability to increase student motivation and develop computational thinking in programming education [26], [27]. In higher education, AI-driven chatbots provide personalized student support, reduce educator workloads, and offer immediate assistance to learners [29], [31]. Despite these advancements, challenges such as system stability, accuracy, and scalability persist. Nonetheless, the integration of generative AI continues to offer significant opportunities for more efficient and personalized education, underscoring its transformative potential [28], [32].

Generative AI tools provide customized support by adapting to individual learning styles and preferences, enhancing educational experiences through personalized assistance, whether via AI-powered tutors recommending relevant study materials or adaptive platforms adjusting content difficulty [27], [33]. Real-time feedback, enabled by automated grading systems, chatbots, and interactive simulations, allows students to promptly correct misconceptions [29], [33]. AI-driven gamification, virtual labs, and interactive quizzes further engage students, making learning more enjoyable and interactive [27], [33]. Additionally, AI chatbots offer a non-judgmental space for inquiries, reducing the social pressure felt by students who may hesitate to ask questions in a traditional classroom due to social anxiety [32]. A recent survey demonstrates that over 98% of higher education students have used generative AI tools for learning support, underscoring the importance of understanding their educational impact [34]. For educators, generative AI automates repetitive duties, allowing more time for impactful activities, and personalized interactions with students [27], [35]. Moreover, generative AI's ability to analyze student performance data provides insights into progress, highlights patterns, and identifies areas for improvement, enabling educators to fine-tune their teaching strategies [25], [35]. Nonetheless, several challenges persist, such as guaranteeing the precision and trustworthiness of responses produced by AI systems [36], [37], and addressing moral issues connected to algorithmic bias, information confidentiality, fairness, and transparency within AI models [36], [38], [39].

As AI-driven education advances, critical ethical and technical challenges arise. A major concern is the inherent bias in AI algorithms, which can inadvertently reinforce inequalities if not carefully managed [36], [38]. AI systems learn from extensive datasets which might include implicit biases, leading to unfair treatment of underrepresented student groups. Addressing fairness in AI-based learning requires transparent algorithmic decision-making and strategies to

mitigate potential biases [39]. Privacy concerns also persist. As AI-powered educational tools collect large volumes of learner information, prompting concerns over information safety and data ownership [36]. Ensuring the responsible use of AI in education necessitates robust privacy frameworks and ethical guidelines that prioritize student well-being.

Moreover, emerging trends in AI-enhanced learning, such as adaptive learning and AI-driven assessments, are reshaping personalized education. Adaptive learning platforms leverage real-time data to continuously adapt the complexity and material of lessons according to each learner's performance [27], [33]. AI-driven assessments offer instant feedback and customized evaluations, allowing students to identify knowledge gaps and improve their understanding in real-time [29], [33]. These innovations provide significant opportunities for enhanced engagement and tailored learning experiences, bridging gaps in traditional educational approaches.

The increasing integration of generative AI in education reflects a broader shift toward adaptive, personalized, and data-driven learning. While the technology presents substantial opportunities for improving both teaching and learning, it also introduces complex ethical, technical, and pedagogical challenges that must be addressed. This paper aims to systematically explore how generative AI is being applied in educational contexts, what benefits and limitations have been identified in the literature, and how these findings can inform future directions. By doing so, it contributes to a deeper understanding of the evolving role of generative AI in shaping the future of education. More specifically, this study offers a structured synthesis of recent empirical and conceptual research, highlighting methodological trends, ethical challenges, and practical implications that can inform future AI-driven educational practices.

The rest of this paper is organized as follows. Section II presents an overview of the related literature, Outlining significant contributions in the field, Section III explains the research methods used in this study. In Section IV, the findings of the analysis are presented, followed by a comprehensive discussion in Section V, which includes the study limitations and recommendations for future investigations. Finally, Section VI provides a summary of the conclusions drawn from the study.

II. RELATED WORK

Over the past few years, generative AI has received considerable attention for its potential to revolutionize educational practices, offering new avenues for assessment, personalized learning, and institutional management [25], [35]. This systematic literature review seeks to synthesize and evaluate the present landscape of research in this dynamic domain. Although earlier research has examined the revolutionary impact of AI in education, this review aims to further investigate the intricate issues and difficulties surrounding the adoption of generative AI within educational contexts. Table 1 provides a comparative analysis with relevant studies,

highlighting the distinctive contributions and gaps addressed by this review.

The research [40] provides a thorough exploration of how generative AI influences methods used in educational assessment practices. the PRISMA framework extension [45] designed for scoping reviews, the study meticulously selected and analyzed 32 empirical studies, highlighting generative AI's transformative effects across students, teachers, and educational institutions. The review particularly emphasizes the role of technologies like ChatGPT in reshaping assessment methodologies and promoting self-regulated learning. However, a significant critique is the limited discussion on ethical considerations, particularly concerning the implementation of ChatGPT alongside other powerful AI tools. The study primarily focuses on ChatGPT, potentially overlooking other equally impactful generative AI technologies.

The study [41] provides valuable insights regarding the convergence of generative AI technologies with English Medium Instruction (EMI) within the context of higher education. It explores current research and offers a roadmap for future investigations. The study highlights generative AI's potential to enhance language acquisition, content development, and evaluation, noting that the field is still emerging but promising. The methodology employs the PRISMA framework [45] to systematically review and select relevant publications. However, the study has significant limitations, including a lack of discussion on data privacy issues related to generative AI's role in transforming all areas of education, not just language acquisition and content development. Additionally, the focus is predominantly on ChatGPT, potentially overlooking other powerful generative AI tools and technologies that could have significant implications.

The study [42] provides a look at the existing studies on generative AI within higher education. It identifies how generative AI is evolving in teaching and learning, highlights its potential and limitations, and suggests areas for future research. Using PRISMA guidelines [45], the study systematically reviews scholarly articles from the past seven years, employing topic modeling and bibliometric analysis to uncover key themes and trends. However, the study has notable limitations. It primarily focuses on Scopus publications, potentially overlooking relevant research from other databases, and offers an overview without deeply exploring the impact of generative AI on different educational aspects. Additionally, it lacks innovative ideas, detailed discussions on data privacy, and broader implications of generative AI in education.

The study [43] come up with meaningful understanding of the effects of generative AI on language education, using a scoping review methodology to outline the current research landscape. It highlights the versatility of generative AI tools across various education levels and languages and identifies key research areas, highlighting the necessity for additional empirical research on the efficacy of generative AI tools, and emphasizes the importance of addressing ethical

TABLE 1. Comparison of related studies.

Paper	Year	Type	Number of papers	Time-span	Source
A scoping review on how generative artificial intelligence transforms assessment in higher education [40]	2024	Journal	32	2018-2023	ERIC, WoS, and Scopus
A systematic review of generative AI and (English medium instruction) higher education [41]	2023	Journal	68	2022-2023	Google Scholar, Scopus, WoS, and ERIC
A Systematic Review of Generative AI for Teaching and Learning Practice [42]	2024	Journal	335	2017-2023	Scopus
Application of generative artificial intelligence in language teaching and learning: A scoping literature review [43]	2024	Journal	41	2017-2023	Google Scholar, Scopus, Science Direct, ERIC, JSTOR
Transforming Education A Comprehensive Review of Generative Artificial Intelligence in Educational Settings through Bibliometric and Content Analysis [44]	2024	Journal	207	2018-2023	Scopus
This study	2024	Journal	55	2017-2024	WOS

considerations, data privacy, and security. The review underscores the role of stakeholder engagement in the implementation of generative AI in language education. However, the study has important limitations, such as its focus on Google Scholar more than Scopus and ScienceDirect, and its lack of discussion on other aspects of learning, such as coding.

The research [44] offers meaningful insights into the integration of generative AI in educational settings. The study uses a systematic method that combines bibliometric and content analysis to examine citation patterns, co-authorships, and keywords. This helps track how research topics evolve and identifies important authors in the field. Through content analysis, the study categorizes and classifies research papers, revealing the diverse applications of generative AI in education, including popular models like ChatGPT and emerging tools such as AlphaCode [46] and GitHub Copilot [47], emphasizing their potential for transforming teaching, learning, and research practices. Furthermore, the study addresses critical issues such as ethical considerations, responsible AI use, data privacy protections, biases in AI systems, along with the need to uphold academic integrity. It underscores the necessity for transparent practices and AI-enhanced curriculum development that aims to respond to the unique learning requirements of students. However, the study’s exclusive focus on the Scopus database restricts its breadth, potentially overlooking significant research available in other databases. Moreover, it lacks innovative strategies to effectively mitigate the identified limitations and enhance the robustness of generative AI applications in educational environment.

Despite the increasing adoption of generative AI in educational settings, there remains an insufficiency of comprehensive understanding regarding its full implications and challenges. While existing literature highlights the potential transformative impact of generative AI on educational practices such as assessment and learning, there is a notable absence of in-depth exploration into the practical implementation complexities and unresolved issues. Moreover, the

rapid evolution of generative AI technologies necessitates ongoing literature reviews to capture recent advancements and emerging trends. This study seeks to bridge this gap by performing a systematic review aimed at clarifying the challenges and opportunities presented by generative AI in education, while addressing key limitations found in previous reviews — including narrow focus on specific tools like ChatGPT [40], [41], limited attention to ethical concerns and data privacy [41], [42], [43], restricted database coverage [43], [44], and lack of practical recommendations or innovative strategies [42], [44]. By responding to these gaps, this research aims to offer meaningful insights to the domain of educational technology and inform both future investigations and applied practices in leveraging generative AI for educational enhancement.

III. RESEARCH METHODOLOGY

A. REVIEW QUESTIONS

To systematically explore and synthesize the current collection of research regarding the adoption of generative AI within the context of higher education, it is essential to establish a clear framework guided by well-defined review questions. Developing well-defined review questions is a critical initial step in performing a systematic literature review, as it determines the direction, scope, and goals of the review process. In this research, we rigorously adhered to established SLR protocols to maintain methodological thoroughness and completeness [48]. We followed a structured and reproducible approach, ranging from establishing inclusion and exclusion criteria to conducting a systematic search and selecting pertinent studies.

To develop the search query, we utilized the Population, Intervention, Comparison, and Outcome (PICO) framework [49], a widely recognized method for structuring research questions and constructing search strategies in systematic reviews. In our case, PICO helped frame the review questions by defining ‘Population’ as the use of generative AI in education, ‘Intervention’ as the AI tools and methods applied, ‘Comparison’ as traditional edu-

cational approaches, and ‘Outcome’ as improvements in learning or teaching effectiveness. These elements directly informed both the formulation of our research questions and the design of the search string, ensuring alignment between the review objectives and the retrieved literature. Additionally, A stringent screening process was applied to select papers that adhered to the predefined inclusion criteria, which encompassed relevance to generative AI in higher education, comprehensive implementation details, and empirical evaluations. This systematic method allowed us to identify and examine a targeted yet representative collection of papers that greatly enhanced our understanding of the subject.

This section outlines the research questions that guide our systematic literature review. By specifying the inquiries we aim to address, These questions not only define the trajectory of the review but also offer readers a comprehensive understanding of the study’s key themes and goals.

- **RQ1:** What are the most popular tools in generative AI used in the realm of education field?
- **RQ2:** How does the use of generative AI tools in education impact student learning outcomes compared to traditional methods?
- **RQ3:** How have studies evaluated the effectiveness of generative AI applications in enhancing educational outcomes, and what methodologies have they used to assess these applications?
- **RQ4:** What are the ethical considerations and limitations related to the application of generative AI within academic environments?
- **RQ5:** How can generative AI enhance the future of education?

Every research question in this SLR contributes significantly to a unified narrative that methodically investigates different dimensions of generative AI in education. Investigating the most popular generative AI tools used in the educational field leads naturally to examining their impact on student learning outcomes, as understanding which tools are prevalent helps in assessing their effectiveness compared to traditional teaching methods. Similarly, evaluating the effectiveness of generative AI applications and the methodologies used in studies is closely linked to comprehending the overall impact of these tools on learning outcomes. Furthermore, exploring the moral implications and limitations, such as data privacy and bias, is inherently connected to the ethical application of generative AI in education, ensuring that these tools are implemented in a way that protects students and educators. Exploring how generative AI can shape the future of education is closely linked to recognizing current trends and anticipating future possibilities, offering a holistic perspective on the dynamic evolution of generative AI within educational contexts. This methodical approach enables us to frame the advantages, challenges, and existing gaps in the literature, thereby informing future research trajectories and priorities concerning the incorporation of generative AI in education.

B. STUDY SELECTION CRITERIA

Defining clear inclusion and exclusion criteria is essential for conducting an SLR on generative AI in education. These criteria establish a structured framework that delineates the scope of our review, allowing us to systematically include relevant studies while filtering out those that do not meet the established standards. This section outlines the specific parameters and rationale underlying our inclusion and exclusion criteria. Well-defined and transparent guidelines uphold the methodological rigor of our review and enhance its reproducibility. These criteria act as a critical sieve, enabling us to filter through the vast body of literature and identify studies that align with the objectives of our SLR. By maintaining a high standard of quality and relevance in selecting primary research sources, we ensure that our findings and conclusions are grounded in a systematically curated and representative body of evidence.

1) DATABASE SELECTION

For this systematic literature review on generative AI in education, we selected the Web of Science (WoS) as our primary database. WoS was chosen for its rigorous indexing standards, comprehensive coverage of high-impact, peer-reviewed journals, and its strong representation of multidisciplinary research. Given that generative AI in education lies at the intersection of computer science, education, and social science, WoS offers a reliable and well-curated source of relevant literature across these domains.

This systematic review also aligns with established guidelines for conducting standalone literature reviews in education and information systems, ensuring methodological transparency and rigor [50].

Moreover, WoS provides advanced search functionalities, including Boolean operators, proximity searching, and detailed filtering options, which enable precise and reproducible search strategies — a critical component of systematic reviews. Its long-standing reputation and widespread use in evidence-based research further support its selection as a trusted and methodologically appropriate source.

While other academic databases such as Scopus, IEEE Xplore, or ScienceDirect also contain valuable content, our decision to focus on WoS was driven by its balance of quality, breadth, and search efficiency. We acknowledge that limiting our search to a single database may exclude some relevant studies indexed elsewhere; however, this approach ensures a high level of quality control and maintains the review’s methodological rigor.

2) SEARCH QUERY

To construct the search query, we employed the PICO criteria [51]. PICO stands for Population, Intervention, Comparison, and Outcomes, and helps to systematically develop search terms from the review questions. In the context of our study:

- **Population:** Application of generative AI in education.

- **Interventions:** generative AI tools, techniques, and methods used in educational settings.
- **Comparison:** Comparative studies between generative AI applications and traditional educational methods.
- **Outcomes:** Enhancements in educational outcomes and future improvements through generative AI.

Based on these criteria, we formulated the following search query, which was conducted at the end of the third quarter of 2024: (“generative AI” OR “genAI” OR “Generative artificial intelligence” OR “Large Language Model” OR “LLM” OR “Generative models” OR “Large Vision Model” OR “LVM”) NEAR/50 (“E-learning” OR “teacher” OR “student” OR “academic” OR “school” OR “university” OR “universities” OR “eLearning” OR “education” OR “online learning” OR “virtual learning” OR “Distance learning” OR “Digital education” OR “Blended learning” OR “Learning management system” OR “Massive open online course” OR “LMS” OR “Open educational resources” OR “OER” OR “digital education”)

We utilized Boolean operators to craft a comprehensive search strategy designed to identify studies related to the application of generative AI in educational settings. Here is a breakdown of the query:

- (“generative AI” OR “genAI” OR “Generative artificial intelligence” OR “Large Language Model” OR “LLM” OR “Generative models” OR “Large Vision Model” OR “LVM”): This segment of the query employs the OR operator to capture studies that refer to generative AI using various terminologies. Enclosing these terms in parentheses ensures that any of these terms will be matched in the search.
- NEAR/50: The NEAR/50 operator combines the two main parts of the query, ensuring that retrieved papers include terms from both segments within 50 words of each other, focusing on generative AI within educational settings.
- (“E-learning” OR “teacher” OR “student” OR “academic” OR “school” OR “university” OR “universities” OR “eLearning” OR “education” OR “online learning” OR “virtual learning” OR “Distance learning” OR “Digital education” OR “Blended learning” OR “Learning management system” OR “Massive open online course” OR “LMS” OR “Open educational resources” OR “OER” OR “digital education”): This section uses the OR operator to search for terms related to education. This inclusive approach ensures that the search captures studies across various educational contexts and terminologies.

By combining these elements, the search query aims to retrieve a comprehensive set of papers discussing the intersection of generative AI and education, covering applications, methodologies, challenges, and future advancements.

3) INCLUSION AND EXCLUSION

After establishing the search query and selecting the databases, we executed the search across WoS platform, yielding a total of 940 papers. We then applied a series of criteria to refine the dataset further:

- **Exclude papers not written in English:** This criterion led to the exclusion of 7 papers.
- **Exclude papers published outside the timespan of 2017-2024:** This criterion resulted in the removal of 47 papers.
- **Exclude duplicated papers:** This step excluded 2 papers.
- **Exclude papers that are not ‘Article’ or ‘Review article’:** This criterion led to the exclusion of 189 papers.
- **Exclude papers whose titles are unrelated to our topic:** This criterion excluded 533 papers.
- **Exclude articles whose do not have an abstract:** This criterion resulted in the exclusion of 4 papers.
- **Exclude papers whose abstracts do not align with our topic:** This criterion resulted in the exclusion of 103 papers.

A flow chart that summarizes the inclusion/exclusion criterion procedures is presented in Figure 1.

C. QUALITY ASSESSMENT

To uphold the rigor and dependability of this SLR, a pivotal step involves thoroughly assessing the quality of the selected studies. This section plays a crucial role, as each included study undergoes meticulous scrutiny to evaluate its methodological robustness and the reliability of its results. Our objective is to provide readers with a transparent and impartial assessment of the included studies, ensuring that our conclusions rest on solid, well-executed research. By employing a systematic and standardized evaluation process, we seek to identify the strengths and limitations of each study, enabling us to appropriately gauge their contributions in relation to our research objectives. This stringent quality assessment bolsters the credibility of this SLR and assists readers and researchers in determining the reliability of the synthesized evidence, thereby strengthening the validity and significance of the findings.

In our quality assessment process, we focused on three key questions to evaluate the primary studies:

- Does the paper actually apply/discuss generative AI tools to enhance educational outcomes?
- the methodology for implementing generative AI in educational settings clearly defined (problem identification, generative AI techniques, solution framework)?
- Is there empirical evidence supporting the effectiveness of the generative AI solution, such as practical case analyses, experimental findings, or assessment results?

Following the application of the inclusion/exclusion criteria and the completion of the quality assessment, we incorporated a total of 55 papers into this SLR. Although this review may seem more focused than broader surveys, it is important

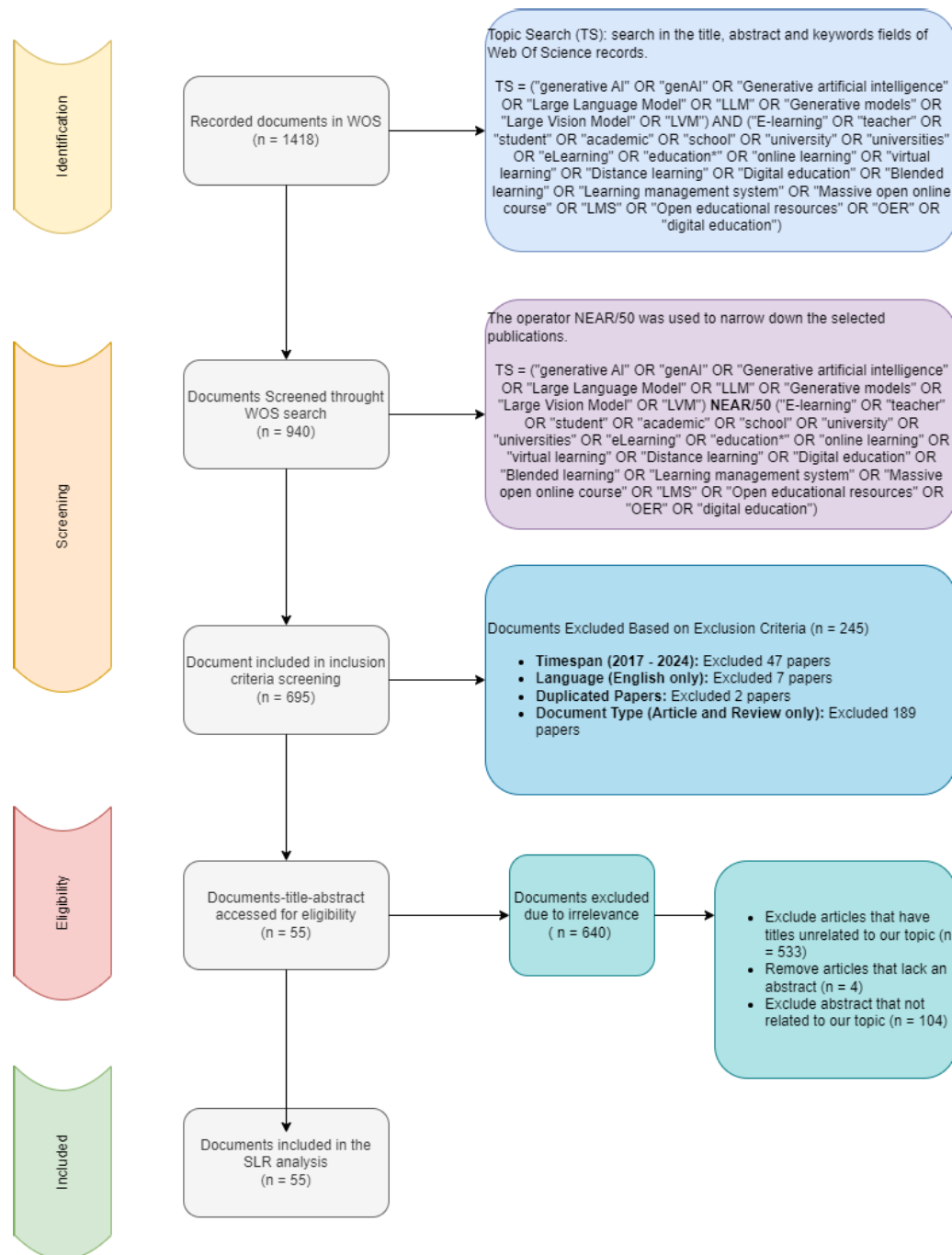


FIGURE 1. Diagram illustrating the inclusion and exclusion criteria process.

to emphasize that our focus was on analyzing a carefully chosen selection of high-quality, full-implementation papers. We ensured that the selected studies offered substantial insights and employed rigorous methodologies that are relevant to generative AI in education. By adhering to these standards, we aimed to maintain the integrity and reliability of our review, thereby establishing a strong foundation for synthesizing key findings and identifying recurring themes and challenges in the field.

D. DESCRIPTIVE KEYWORD ANALYSIS

As part of our methodological approach, we included a descriptive keyword analysis to enhance transparency and contextual understanding of the reviewed literature. By examining the most frequent terms within the titles and abstracts of the included studies. This visualization 2 serves as a complementary perspective to our formal inclusion criteria, highlighting prevalent topics such as personalized learning, AI-assisted teaching tools, and student engagement.

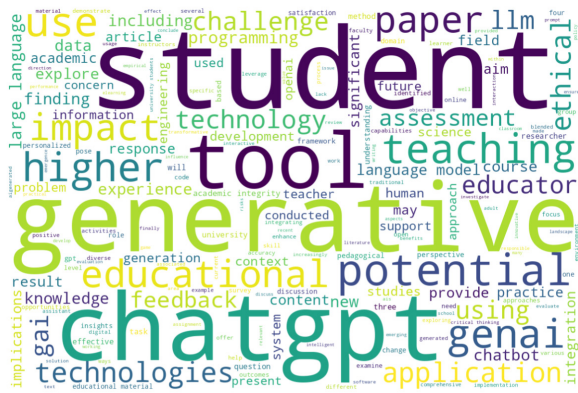


FIGURE 2. Keyword frequency visualization generated from the titles and abstracts of the reviewed studies, illustrating dominant research themes in generative AI and education.

Including this keyword analysis supports a more comprehensive understanding of the landscape of generative AI in education and helps identify focal points that have shaped recent research directions.

E. FINAL PAPERS DATASET

As illustrated in Figure 3, the use of generative AI in education has seen a significant rise in research activity, particularly in recent years. The initial years of the review period, from 2017 to 2021, exhibited little to no research output in this field, with no selected papers identified during this time. However, a single paper was found in 2022, indicating the beginning of interest in this area. 2023 marked a turning point, with a substantial increase in interest, resulting in the selection of 19 papers, reflecting a growing recognition of the potential of generative AI in education. This upward trend continued into 2024, with an even more impressive selection of 35 papers. This substantial growth underscores the expanding focus on exploring and integrating generative AI tools and methodologies in educational settings, highlighting the evolving landscape of this research area.

IV. RESULTS

A. RQ1: WHAT ARE THE MOST POPULAR TOOLS IN GENERATIVE AI USED IN THE REALM OF EDUCATION FIELD?

The integration of generative AI tools into education has revolutionized how educators and students interact with content, streamline administrative tasks, and personalize learning experiences. Through an analysis of selected studies, certain tools have emerged as particularly influential in the educational landscape. These technologies are known for their strength, adaptability, and capacity to cater to various educational needs.

1) OPENAI'S GPT SERIES

The GPT models developed by OpenAI, notably GPT-3 and GPT-4 [52], have become a cornerstone in educational

applications due to their exceptional capability in producing text that closely resembles human language and their versatility in addressing a broad spectrum of academic needs [53], [54], [55]. These models are extensively used for automated essay grading, creating personalized study materials, and providing real-time tutoring [56], [57]. Their adaptability is evident in their capacity to simulate dialogues for language learning, offer personalized feedback on assignments, and even support programming education through code generation and correction [56], [57], [58], [59]. The effectiveness of OpenAI models in producing contextually accurate and relevant content has been highlighted across multiple studies, solidifying their role as a cornerstone in generative AI applications in education [59], [60].

2) DALL-E AND SIMILAR VISUAL GENERATIVE MODELS

In addition to text-based tools, visual generative models such as DALL-E and Midjourney [60], [61] have gained popularity in educational environments that emphasize creativity and design [54]. These tools are employed to generate images from text descriptions, which can be especially beneficial in disciplines such as art, history, and science, where visual aids are crucial for facilitating comprehension. The capacity of these models to create bespoke visual content tailored to specific educational requirements has been identified as a noteworthy advantage, particularly in contexts where resources are limited and access to diverse visual materials may be constrained [22], [61].

3) GEMINI AND BING AI

Gemini and Bing AI [60], [62], developed by Google and Microsoft respectively, are emerging as key tools in education, particularly for information retrieval, content generation, and interactive learning [60], [63], [64]. Gemini uses Google's extensive data to provide detailed, contextually relevant responses, making it a valuable resource for research and independent learning [53], [55], [65]. Bing AI combines AI-driven content creation with search functionalities, effectively generating educational materials like summaries, explanations, and study guides. These tools enhance classroom discussions and support educators by automating content creation, and their presence in education is anticipated to grow, providing increasingly personalized support for student learning and teaching [53], [62].

Overall, the analysis of the most popular generative AI tools in education reveals a strong preference for those that offer flexibility, ease of integration, and the ability to personalize learning experiences. However, it is important to recognize that most reviewed studies focused on widely known, English-centric tools. Less-publicized and non-English generative AI systems remain underexplored, limiting insights into their potential to support diverse linguistic and cultural educational needs. As these tools continue to evolve, their impact on education is likely to grow,

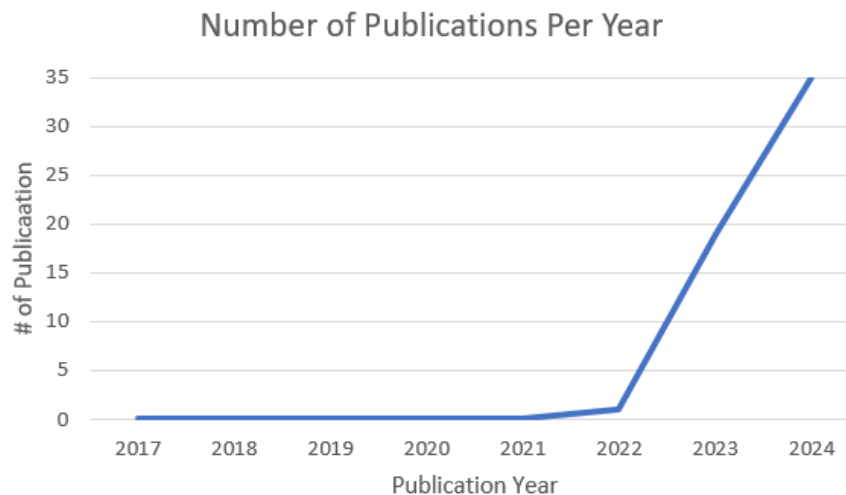


FIGURE 3. Field growth since 2017, showing the increase in research publications related to generative AI in education.

paving the way for more innovative and effective teaching and learning strategies.

B. RQ2: HOW DOES THE USE OF GENERATIVE AI TOOLS IN EDUCATION IMPACT STUDENT LEARNING OUTCOMES COMPARED TO TRADITIONAL METHODS?

To understand the influence of generative AI technologies on learners' academic performance, it is essential to first compare such technologies with traditional educational methods. While traditional models have long been the standard in education, the introduction of generative AI presents new possibilities for enhancing the learning experience. This section explores how generative AI tools influence student outcomes by offering personalized learning, encouraging creativity, and fostering engagement, compared to the more rigid and uniform approaches of traditional education.

In traditional educational settings, standardized curricula and uniform teaching methods often dominate, which can overlook the different ways students learn and their individual strengths and interests. This rigid approach can lead to disengagement and limit opportunities for personal growth, as students are expected to fit into a one-size-fits-all model that might not suit their unique needs [14], [15], [16]. Additionally, traditional education is frequently criticized for focusing too much on memorizing facts rather than on developing important skills like critical thinking, problem-solving, and creativity [17], [18]. When education is centered around the teacher as the main source of knowledge, it can reduce students' ability to think for themselves and engage deeply with the material [19]. This teacher-centered approach also makes it harder for students to become independent thinkers, as they are often expected to passively receive information rather than actively participate in their learning [66]. The rigid structure of traditional education leaves little room for adapting to individual needs, making the educational experience more about meeting set standards

than encouraging a genuine love of learning [19]. Moreover, the pressure of standardized testing can increase stress and anxiety among students, taking away from the overall learning experience [14], [17].

Generative AI holds significant promise for addressing the limitations of traditional educational models by offering innovative possibilities for teaching and learning [67], [68]. By utilizing generative AI, teachers are able to design increasingly flexible, customized, and engaging educational settings [23], [69]. One of the key advantages of generative AI is its ability to personalize learning experiences [23], [24]. These AI systems can analyze vast amounts of data on individual students' learning preferences, progress, and challenges, tailoring content and feedback to meet their unique needs [70]. This level of personalization helps ensure ensuring every learner gets the assistance they need to succeed, thereby enhancing their overall learning experience [24], [71].

In addition to personalization, generative AI brings interactivity into the classroom through AI-powered tools and virtual assistants [22]. These tools can handle and simulate complex real-world scenarios to produce a human-like output [23], provide instant feedback, and offer dynamic, hands-on learning experiences that keep students engaged [22], [24], [72]. Furthermore, the use of generative AI in creating adaptive learning pathways allows students to explore topics at their own pace, ensuring that faster learners are continuously challenged while others receive the additional support they need [73]. These AI-driven assessments can analyze student responses in depth, identifying areas of strength and weakness, and adjusting future learning activities accordingly [74]. This ensures that assessments are more reflective of individual learning journeys and more conducive to meaningful educational outcomes.

Moreover, generative AI can potentially significantly reduce the administrative burden on educators. Through

the automation of tasks like data analysis [74], AI allows teachers to focus more on instructional strategies and student interaction [70], [75]. Artificial intelligence applications are able to examine student performance data to identify trends and recommend appropriate actions, assisting instructors in providing targeted support where it is most needed [75], [76]. This improves teaching efficiency and helps educators address diverse student needs [23], [77].

A systematic review has found that ChatGPT can boost student engagement and accessibility, while also highlighting critical concerns about plagiarism and accuracy that must be managed thoughtfully [78].

Generative AI is also instrumental in fostering creativity and critical thinking among students [79]. It can generate new content, ideas, and solutions, encouraging students to think outside the box and explore innovative approaches to problem-solving [23], [59]. This is particularly valuable in project-based learning environments, where AI can assist in brainstorming, research, and content creation, allowing students to delve deeper into subjects and develop a more profound understanding [80].

In short, the rise of generative AI in education represents a transformative shift in how teaching and learning are approached. By offering personalized, interactive, and flexible learning experiences, along with innovative assessment and support tools, generative AI has the potential to overcome many of the limitations inherent in traditional educational models. As educators continue to integrate AI into their practices, the educational landscape will likely evolve to become more inclusive, engaging, and effective. Teacher and student perceptions also play a crucial role in the successful integration of generative AI in education. Students generally appreciate the immediacy, personalization, and flexibility that AI tools offer, while teachers often express both interest and caution—particularly in relation to accuracy, monitoring student use, and time required for oversight. These perceptions directly influence adoption, implementation, and sustainability of AI-enhanced learning environments.

C. RQ3: HOW HAVE STUDIES EVALUATED THE EFFECTIVENESS OF GENERATIVE AI APPLICATIONS IN ENHANCING EDUCATIONAL OUTCOMES, AND WHAT METHODOLOGIES HAVE THEY USED TO ASSESS THESE APPLICATIONS?

Integrating Generative AI in education has sparked significant interest, with various studies evaluating its effectiveness in enhancing educational outcomes. This research question seeks to explore how these studies have assessed the impact of generative AI applications in education and the methodologies employed in their evaluation. By analyzing multiple research works, a comprehensive understanding of the role and effectiveness of generative AI in educational settings can be derived. Table 2 presents a comparative analysis of selected studies, highlighting their objectives, methodologies, AI technologies used applications in education, key findings, benefits, and challenges.

Table 2 provides a comprehensive comparison of studies that evaluate how effective generative AI applications are in improving learning results, illustrating the diverse methodologies used to assess these technologies. The studies explore a variety of educational contexts, from academic writing in English as a Foreign Language (EFL) and higher education research [26] to literature reviews and coding tasks in public policy education [84]. The methodologies employed in these studies range from qualitative approaches, such as thematic and content analysis of student essays and interviews [83], to quantitative methods, including quasi-experimental designs [26], empirical field experiments [81], and surveys [79], [82]. This methodological diversity enables a detailed examination into the ways generative AI affects various aspects of education. In terms of application, generative AI tools like ChatGPT have been integrated into various educational activities [55], [79], [81], [83], [84], including writing assignments, literature reviews, and academic tasks. The studies consistently show that these tools can enhance students' critical thinking, improve efficiency, and foster personalized learning experiences [55], [84]. For instance, the use of generative AI in academic writing has led to significant improvements in writing skills [26], particularly when combined with regular feedback. Similarly, the integration of AI in literature reviews and research tasks has been found to streamline processes and improve workflow efficiency, particularly under time constraints. However, the studies also highlight several issues linked to integrating generative AI within educational contexts. Ethical concerns [82], [84], including concerns about the openness and possible bias present in content produced by AI systems [82], are a significant issue. Another concern is the possibility that learners might depend too heavily on AI, which in turn may impede the growth of analytical thinking skills and reduce their autonomy throughout their educational journey. Additionally, the accuracy and validity of AI-generated content pose challenges, particularly in maintaining academic integrity and avoiding plagiarism [82]. These concerns are compounded by the difficulties students face in verifying AI-generated information, which can further complicate the integration of these technologies into educational settings [84]. The studies also reveal that the effectiveness of generative AI tools can vary based on individual differences among students, such as their initial attitudes towards AI, task motivation, and personal traits [79], [81], [83]. This variability underscores the importance of context-specific research and the need for adaptive educational strategies that consider these differences. Moreover, although generative AI tools hold strong promise for improving educational outcomes, their successful integration into education demands thoughtful attention to these concerns, as well as ongoing support for educators and learners as they deal with the moral and real-world consequences of implementing AI.

Recent experimental and applied studies have provided additional quantitative evidence on the effectiveness of

TABLE 2. Comparative review of research on generative AI in educational contexts.

Authors and Year	Objective	Methodology	AI Technology	Application	Results/Findings	Benefits	Challenges
Ayse Taskiran and Nil Goksel - 2022 [26]	Investigate the effects of both automated and instructor feedback on academic writing performance in EFL contexts within open and distance learning environments	Quasi-experimental, quantitative, repeated measures, weekly tasks over nine weeks, feedback from teachers (first 3 tasks) and software (last 3 tasks), diagnostic test, and post-tests	AI-based automated feedback software (specific technology not detailed)	Improving academic writing in EFL, open and distant learning context	Significant improvement in writing skills with regular feedback, higher improvement with teacher feedback	Enhances writing skills, reduces mechanical errors, supports lifelong learning, makes learning more interesting and joyful	Small sample size (36), need for larger groups, suggest qualitative designs and diverse methods
Wang, Jiyao and Huang, Chunxi and Yan, Song and Xie, Weiyin and He, Dengbo - 2024 [81]	The primary objective of this research was to assess the performance of early-stage scholars using large language models (LLMs) for academic tasks, particularly focusing on how personal variability and the complexity of tasks affect outcomes	The approach involved an empirical field experiment with 48 early-stage scholars. They performed paper understanding and literature review under varying time constraints, after receiving different training programs on LLM capabilities and limitations	ChatGPT	The context of the research pertains to academic tasks, focusing on paper reading and literature reviews, which are integral activities in academic work	Young scholars generally maintained high academic integrity while using LLMs. Performance varied by task type and time pressure, but not by training type. Scholars' traits also influenced their performance and attitudes towards LLMs	LLMs can alleviate time pressure during academic tasks, suggesting potential improvements in workflow efficiency for scholars	LLMs face challenges like inadequate synthesis, plagiarism risks, and difficulty producing publishable quality writing. They also struggle to meet the rigorous standards of academic tasks for logical consistency, accuracy, and originality.
Abdulrahman M. Al-Zahrani - 2023 [82]	To explore the perceptions and utilization of generative AI tools among higher education students, focusing on awareness, impact, usage, institutional support, ethical considerations, and future outlooks in research practices	A survey methodology was employed, involving 505 higher education students in Saudi Arabia. Data were analyzed using descriptive statistics and Multivariate Analysis of Variance (MANOVA) with (Statistical Package for the Social Sciences) SPSS v-22	generative AI tools, No specific models were named	The study examined the integration of generative AI tools in research within higher education, enhancing research productivity, facilitating collaboration, and improving research outcomes	The study revealed a high level of awareness and positive perceptions of generative AI tools among participants, who recognized their potential to revolutionize academic research by improving efficiency and expanding research scope	Improve research efficiency, expanded research scope, enhanced collaboration, and reduced time for data analysis. Participants expressed confidence in the effective use of these tools	Ethical concerns include transparency, potential biases, inaccuracies affecting result validity, and issues related to academic integrity and plagiarism
Matt Bower, Jodie Torrington, Jennifer W. M. Lai, Peter Petocz, and Mark Alfano - 2024 [79]	To examine educator perceptions of how teaching and assessment should evolve in response to generative AI technologies like ChatGPT, and to understand the motivations behind these potential changes	A mixed methods approach was used, surveying 318 educators from various regions, disciplines, and levels of experience. Both qualitative and quantitative data were collected to assess beliefs regarding the impact of generative AI on teaching and assessment	Focused on generative AI technologies, particularly ChatGPT	Adapting curriculum, pedagogy, and assessment methods to incorporate AI, with an emphasis on promoting higher-order thinking and embedding ethical values in education	A majority of educators believe generative AI will significantly impact teaching and assessment, with variations in belief observed across different teaching levels, disciplines, regions, and genders. Many educators acknowledge the need for change	Enhanced performance expectancy for both teachers and students, improved learning processes, and deeper student engagement with educational material	self-selection bias, a limited sample size that is insufficient to generalize the results of the findings. Additionally, there is a critical need for context-specific research and understanding the professional learning needs of teachers to effectively integrate AI in education.
Mohammad Hmoud, Hadeel Swait, Nardin Hamad, Omar Karram, and Wajeeh Daher - 2024. [83]	To investigate higher education students' task motivation when using ChatGPT for assignment completion, focusing on autonomy, competence, and relatedness	Qualitative content analysis was conducted, with thematic analysis of interviews from 15 graduate students who had experience using ChatGPT. Data was analyzed using MAXQDA 2022	GPT3.5	Utilized for assisting students with university assignments, showing its potential to enhance learning experiences by improving task motivation	Students displayed increased task motivation across five main categories: Task Enjoyment, Reported Effort, Result Assessment, Perceived Relevance, and Interaction, each with detailed sub-categories	Promotes student motivation and academic achievement by creating more engaging and interactive learning experiences	Challenges include potential issues related to students' anxiety and the need for clear feedback mechanisms, as well as differentiating AI-generated work from original student output
Gleb Papyshev - 2023 [84]	To explore the usage of generative AI technologies among research postgraduate students in public policy, focusing on their sentiments and the implications for teaching and learning	Qualitative analysis was conducted through 14 structured interviews with research postgraduate students at the Hong Kong University of Science and Technology (HKUST). Sentiment analysis was also applied to the interview transcripts to capture the range of emotions and perspectives	generative AI technologies were discussed, particularly large language models (LLMs) like ChatGPT, though specific models were not explicitly named	Used for literature reviews, coding tasks, and writing processes within public policy education, demonstrating its utility in streamlining various research activities	The study found a range of sentiments among students regarding the use of generative AI, highlighting both positive experiences (e.g., efficiency in tasks) and negative experiences (e.g., achieving desired outcomes and ethical concerns)	The study highlights potential advantages of generative AI in educational settings, such as enhancing personalized learning, providing continuous feedback, and improving specific tasks like literature reviews	Reported limitations include concerns about academic integrity, the potential for generative AI to be perceived as the ultimate epistemic authority leading to unquestioned acceptance of AI-generated information, and risks associated with decreased creativity and critical thinking skills among students
Josef Sedlbauer, Jan Činčera, Martin Slavík, and Adéla Hartlová - 2024. [55]	The primary aim of the research was to examine undergraduate students' experiences with ChatGPT and to uncover the challenges linked to the growing use of artificial intelligence in educational settings	The study employed a qualitative design, using the general inductive approach to analyze essays written by 25 junior university students. The analysis focused on students' reflections on their interaction with ChatGPT, identifying and categorizing data segments related to their perspectives on AI	GPT3.5	ChatGPT was utilized to assist students in writing seminar essays based on selected scientific literature, integrating AI into their research and writing processes	The findings revealed a range of initial attitudes towards ChatGPT, from enthusiasm to indifference. Over time, most students embraced the technology, noting improvements in critical thinking, strategic competence, and system thinking. However, some challenges were also identified, including difficulties in verifying sources and initial imperfections in AI responses	The use of ChatGPT led to enhanced critical thinking, strategic competence, and personalized learning experiences. Students perceived AI as a dialogical partner, which contributed to better conceptual understanding and the development of problem-solving and communication skills	Reported limitations included a small sample size (N = 25), the specific demographic of university students, and the qualitative nature of the study, which restricts generalization of findings. Additionally, students faced challenges in verifying sources and encountered initial imperfections in the AI's responses.

generative AI in education. For example, randomized controlled trials have demonstrated that AI-feedback systems can significantly improve essay organization and content development compared to instructor feedback [85]. Conversely, controlled experiments have found that ChatGPT-assisted writing can reduce students' cognitive engagement and deep processing, indicating potential drawbacks for critical thinking [86]. Other applied research in authentic educational settings has shown high levels of perceived usefulness, with students reporting enhanced understanding and skill development when integrating ChatGPT into coursework [87]. These diverse results highlight the complex, context-dependent impact of generative AI on learning outcomes, underscoring the importance of rigorous empirical evaluation in future research.

Overall, this comparative review of the studies shows that although generative AI holds the promise to transform educational practices and enhance learning outcomes, it also introduces notable challenges that require careful attention to ensure its ethical and effective application. The range of methodologies used across these studies provides meaningful insights into the intricate relationship between AI technologies and educational results, emphasizing the necessity for robust evaluation frameworks capable of capturing the varied and layered effects of generative AI within the education sector.

D. RQ4: WHAT ARE THE ETHICAL CONSIDERATIONS AND LIMITATIONS RELATED TO THE APPLICATION OF GENERATIVE AI WITHIN ACADEMIC ENVIRONMENTS?

The rise of generative AI in educational environments has unlocked innovative possibilities for improving teaching and learning practices. However, implementing these technologies is not without its difficulties. As with any influential tool, the use of generative AI in education involves a range of ethical considerations and limitations that must be addressed with care. These issues cover various dimensions, including academic integrity, the risk of bias in AI-generated materials, data security concerns, and the persistence of the digital divide. This section explores the ethical challenges and constraints tied to generative AI in academic settings, emphasizing the importance of deliberate and ethical implementation of such technologies to ensure they have a meaningful and beneficial impact on the education system. One of the primary ethical concerns revolves around academic integrity. The ability of generative AI to produce fluent and context-aware text poses a risk of facilitating plagiarism and undermining the authenticity of students' work [88], [89], [90], [91]. This concern is compounded by the potential for students to over-rely on AI-generated content [79], [92], which could diminish their critical thinking and creativity [23], [79], [93]. As students become more dependent on AI tools for writing essays, conducting research, and completing assignments, there is a growing risk that they may disengage from the learning process, resulting

in limited comprehension of the topic [94]. The potential for bias within AI algorithms represents another critical ethical consideration [94], [95]. Generative AI systems rely on massive datasets, which may include underlying biases [93], [96]. When these biases are inadvertently incorporated into educational content, they can perpetuate stereotypes and reinforce discriminatory practices [79]. The absence of clarity in how AI systems make decisions further exacerbates this problem, as students and educators may unknowingly rely on biased outputs, leading to skewed perspectives and misinformed conclusions [91], [94]. Additionally, data privacy is also a critical ethical concern associated with using generative AI in education [73], [97]. The deployment of AI tools often involves collecting and analyzing vast amounts of personal data from students and educators [79], [91], [96]. This data may include sensitive information, such as academic performance, learning preferences, and behavioral patterns [73], [91]. The potential misuse or unauthorized access to this data poses a significant threat to individuals' privacy and could lead to harmful consequences, such as identity theft or targeted discrimination [79], [96]. Educational institutions must establish robust data protection policies and practices to safeguard students' privacy and ensure that AI tools are used responsibly [91], [98]. Finally, it's important to consider the ethical implications of using AI in educational assessments [88], [99]. As generative AI becomes more common in evaluating student work, there's a risk that AI-driven assessments might miss the subtlety and context that human educators can provide [91], [94]. This could lead to unfair or inaccurate evaluations, especially for students whose work doesn't fit the patterns the AI expects [23]. Additionally, the integration of generative AI raises concerns about cheating, as learners could rely on AI systems to produce content that does not reflect their effort, making it challenging to assess their true understanding and abilities [89], [99], [100].

Another crucial ethical consideration involves the trustworthiness of AI-generated outputs in education. Large Language Models can produce highly convincing but factually incorrect or fabricated content, a problem known as hallucination [101]. This raises serious concerns about students unknowingly learning incorrect information or relying on flawed responses in critical contexts [102]. For example, [103] highlight the need for educators to teach critical evaluation of AI outputs, given the risk of hallucinations and misleading content in tools like ChatGPT. Additionally, recent research emphasizes that accepting AI-generated text without thorough validation can undermine academic integrity and trust in educational contexts. Reference [104] found that students who over-rely on tools like ChatGPT risk incorporating inaccurate or even fabricated content into their work, threatening academic ethics and highlighting the need for clear guidelines and training to support fact-checking and responsible use. Similarly, [105] conducted a systematic review showing that over-reliance on AI dialogue systems can weaken critical thinking and problem-solving skills.

They stress the importance of human oversight, teaching students to critically evaluate AI-generated content to prevent them from being misled by plausible but untrue outputs. Addressing these challenges is essential to ensure that generative AI supports, rather than undermines, educational quality and student learning.

E. RQ5: HOW CAN GENERATIVE AI ENHANCE THE FUTURE OF EDUCATION?

With the ongoing evolution of education, generative AI is poised to significantly reshape the future of education. Thanks to its capacity to process and produce large-scale data, generative AI offers innovative solutions that extend beyond traditional methods, addressing challenges and creating new opportunities for personalized learning, accessibility, and curriculum development. This question explores how generative AI can be harnessed to enhance educational outcomes, paving the way for a more effective and inclusive educational system.

Generative AI holds immense potential to revolutionize the future of education by offering transformative solutions that address existing challenges and unlock new opportunities for learning [44], [106]. One of the most significant contributions of generative AI is in the realm of personalized learning [55], [107]. Through the analysis of large-scale data, generative AI can personalize educational content to align with the distinct needs and learning preferences of each student, offering tailored learning experiences that adjust to individual pace and development [53], [106]. This level of personalization can help close achievement gaps, ensure that all students are adequately supported, and foster a more inclusive learning environment [108]. Furthermore, generative AI can significantly enhance curriculum development [89]. By generating diverse educational materials and resources, including lesson plans, interactive content, and assessment tools, generative AI can assist educators in creating more engaging and effective learning experiences [53], [64], [89]. This saves teachers time and allows them to focus on more strategic aspects of education, such as fostering critical thinking, creativity, and collaboration among students [44]. In addition to supporting students and teachers, generative AI can play a pivotal role in making education more accessible [89], [109]. By offering real-time language translation, translation services, and adaptive learning tools, generative AI can break down barriers for students with disabilities, language learners, and those in remote or underserved areas [44], [53]. This democratization of education ensures that quality learning opportunities are available to all, regardless of geographical or socio-economic constraints [96]. The future of education with generative AI also includes a reimagined role for teachers [79]. Instead of replacing educators, generative AI can serve as an invaluable tool that augments their capabilities, allowing them to act as learning facilitators rather than mere information transmitters [108]. By handling routine tasks and providing data-driven insights, generative

AI can free up teachers' time to focus on mentorship, student engagement, and developing higher-order thinking skills [23], [73], [110]. Similarly, [111] highlight that while generative AI offers valuable instructional support, effective integration in education depends on addressing challenges such as inconsistent feedback quality and ensuring comprehensive teacher training. Moreover, generative AI can enhance the assessment process, offering real-time feedback and insights that help students understand their strengths and areas for improvement [112]. This continuous feedback loop can lead to more effective learning outcomes and help students take ownership of their educational journeys [22], [24], [84]. However, it is crucial to ensure that these AI-driven assessments are used ethically and in conjunction with human oversight to maintain fairness and accuracy [79], [88].

Beyond theoretical possibilities, recent pilot studies and classroom implementations demonstrate how generative AI is already enhancing education in practice. For example, [113] used ChatGPT in primary school lessons to adapt learning materials to different knowledge levels, improving engagement and personalization. Reference [114] integrated ChatGPT and DALL-E into middle school history classes, finding higher achievement and motivation compared to traditional teaching. In higher education, studies have examined ChatGPT as a writing assistant, with students reporting better organization and improved drafts in lab reports and essays [115], [116]. Similarly, [117] documented a business course pilot where students used ChatGPT strategically to support language learning and critical thinking. These real-world examples underscore the practical value of generative AI in diverse educational contexts while highlighting the importance of teacher guidance, AI literacy, and critical evaluation to ensure effective, ethical use in the classroom.

V. DISCUSSION

This review consolidates insights from 55 research studies focused on the use of generative artificial intelligence (generative AI) in educational contexts. Echoing broader developments in AI research, the majority of these studies were published during the past two years (2023–2024), highlighting the rapidly growing interest in applying generative AI within learning environments. This increase in scholarly output aligns with advancements in AI technologies and the accelerated shift toward digital education, particularly following the global COVID-19 pandemic. Collectively, these trends show growing recognition of generative AI's potential to improve learning, personalize instruction, and help educators streamline administrative tasks.

The majority of the studies analyzed were sourced from the Web of Science, a database known for its rigorous selection of peer-reviewed articles. This ensured a high level of academic rigor in the reviewed papers, a crucial factor given the multidisciplinary nature of generative AI applications in education, where both technological and pedagogical perspectives must be balanced.

Among the tools discussed, Chat Generative Pre-trained Transformer (ChatGPT) and DALL-E emerged as the most frequently studied. Their prominence in the literature aligns with their widespread popularity and ease of integration into educational contexts. These tools were primarily utilized for automated tasks such as essay grading, content creation, and language learning. Their ability to generate human-like text and images offers educators novel ways to streamline routine tasks, which in turn frees up time for more personalized interactions with students. While this concentration on well-known tools is understandable, it highlights the need for further exploration of emerging generative AI technologies. These include enhanced multimodal interaction, improved domain-specific reasoning, culturally responsive content generation, and more advanced personalization tailored to individual learners' needs. Future research could explore how these tools support subject-specific instruction, promote inclusive education, and adapt content dynamically based on student performance, preferences, and learning styles.

In terms of outcomes, many studies reported encouraging results related to student learning and engagement. For instance, artificial intelligence-driven personalized feedback was shown to enhance student motivation by providing timely and individualized responses, an advantage over traditional methods of instruction. Such findings are consistent with broader trends in adaptive learning technologies. However, the degree of improvement often varied depending on factors such as the complexity of the generative AI tool, the educational context, and the subject matter. Notably, artificial intelligence-driven tutoring systems and adaptive learning platforms were found to have a significant positive impact on student engagement and comprehension. Despite these encouraging results, there is still a lack of research examining the long-term effects of generative AI on academic performance, raising questions about the sustainability of these benefits.

Although the literature presents the potential of generative AI positively, many of the reviewed studies lacked concrete, real-world evidence. There is a notable scarcity of detailed case studies or large-scale implementations demonstrating its effectiveness in authentic educational settings. Future research should explore real-world use cases that evaluate the impact of generative AI on specific learning outcomes. Furthermore, because this review relied exclusively on the Web of Science database, some relevant studies indexed in Scopus, IEEE Xplore, or Google Scholar may not have been captured. This limitation may slightly narrow the scope of the included literature.

In addition to pedagogical impacts, ethical concerns were a recurring theme across many of the reviewed studies. Issues such as academic integrity, including the potential for cheating and the generation of plagiarized content, were highlighted as major challenges. These concerns were particularly acute in cases where students had unrestricted access to generative AI tools during assessments. Although

some studies suggested solutions such as the development of artificial intelligence detection systems or restricting generative AI usage during examinations, these proposals remain largely speculative and require further empirical validation. Beyond academic integrity, concerns regarding biases embedded in artificial intelligence-generated content also emerged. Since many generative AI models are trained on datasets that may reflect historical or societal biases, their deployment in educational settings could unintentionally perpetuate stereotypes or spread misinformation.

From a methodological perspective, the studies reviewed employed a range of approaches, with many focusing on qualitative methods, such as surveys and interviews with students and educators. These approaches offered valuable perspectives on how generative AI tools affected classroom dynamics and student motivation. However, a relatively small number of studies employed more rigorous empirical methods, such as controlled experiments or longitudinal designs, to assess the actual impact of generative AI on educational outcomes. This lack of strong empirical evidence is a common challenge in educational technology research, where early enthusiasm can outpace careful, data-driven evaluation.

Furthermore, the limitations of these methodologies, such as the reliance on self-reported data from students and educators and the lack of long-term controlled studies, must be acknowledged. These constraints may affect the generalizability of findings, as short-term improvements in engagement or comprehension do not necessarily translate into sustained academic success. Future research should prioritize mixed-methods approaches that combine qualitative insights with quantitative analysis, enabling more comprehensive evaluations of generative AI's long-term effectiveness.

Another notable gap in the literature is the absence of studies addressing multilingual applications of generative AI. All of the reviewed studies focused on tools designed for English-speaking learners, despite the increasing linguistic diversity in many educational systems. This narrow focus raises questions about the generalizability of current findings to non-English-speaking environments. Future research should expand to include multilingual and cross-cultural applications of generative AI to better serve a global student population and ensure inclusivity in artificial intelligence-enhanced education.

Looking ahead, several promising avenues for future research have emerged. One particularly exciting area is the potential integration of generative AI with adaptive learning platforms, which could customize lesson plans and exercises based on each student's unique progress and learning style. Such integration would enable more dynamic and personalized learning experiences, allowing students to progress at their own pace while receiving targeted support. For example, combining generative AI-driven personalized learning with immersive environments could offer more interactive and engaging educational content.

In addition, future research should explore the perspectives of key stakeholders, including educators, students, institutional administrators, and policymakers, to ensure that artificial intelligence implementations in education align with real-world needs. Understanding teachers' attitudes toward generative AI, their readiness to incorporate these technologies, and their concerns regarding workload, student engagement, and assessment integrity will be crucial for effective adoption.

Moreover, while generative AI presents significant opportunities, its successful implementation requires overcoming practical barriers such as teacher training, institutional support, and resistance to technological change. Educational institutions must invest in training initiatives to help educators acquire the essential competencies needed to adopt AI-based technologies effectively. Without adequate training, the adoption of generative AI may lead to inconsistent usage, further exacerbating existing disparities in educational access and quality.

Developing clear ethical frameworks and institutional policies regarding generative AI use in education will be essential for addressing concerns around bias, privacy, and academic integrity. Policymakers and educational leaders should collaborate with artificial intelligence researchers to establish guidelines that ensure ethical and responsible use of these technologies in classrooms. Additionally, it is important to recognize that integrating generative AI in education is a global challenge and opportunity. Educational systems around the world vary in language, culture, technological infrastructure, and pedagogical traditions. Research and policy must account for these differences to ensure that AI tools are inclusive, culturally sensitive, and accessible to all learners. Addressing global issues such as linguistic diversity, equitable access in low-resource settings, and context-specific pedagogical practices will be essential for realizing the full potential of AI-enhanced education. Future work should prioritize cross-cultural studies and international collaboration to ensure generative AI serves diverse educational communities effectively and ethically.

VI. THREATS TO VALIDITY

Although this systematic literature review followed rigorous procedures to ensure methodological quality, several potential threats to validity should be acknowledged. First, this review was limited to the Web of Science (WoS) database. WoS is widely recognized for its rigorous indexing and includes a large number of journals and conference proceedings from publishers such as IEEE, Springer, and Elsevier. However, relying on WoS alone may have excluded some relevant studies available only in other databases (e.g., Scopus, ERIC, IEEE Xplore). This constraint may have led to a narrower disciplinary scope, particularly in capturing diverse international perspectives or emerging research published in regional or domain-specific journals. Second, the inclusion criteria favored studies with explicit references to generative AI tools, potentially overlooking broader

or emerging AI applications not labeled under the same terminology. Third, despite manual screening efforts, there remains a risk of subjective interpretation when categorizing and analyzing findings across diverse studies. Lastly, the rapid evolution of generative AI technologies means that newer developments might not have been captured within the review's cutoff period (Q3 2024), potentially affecting the completeness of the findings. These limitations should be considered when interpreting the results and conclusions of this review.

VII. CONCLUSION

In conclusion, this systematic literature review has provided a comprehensive analysis of the current landscape of generative AI in education. By adhering to clear inclusion and exclusion criteria, we ensured the reliability and replicability of our review, addressing challenges such as search term limitations and database constraints through careful keyword selection and manual verification processes. The findings highlight significant advancements in the integration of generative AI tools within educational settings, including their applications in personalized learning and content generation, which reflect the potential of these technologies to reshape traditional educational practices.

However, certain limitations emerged from the review. A notable gap lies in the lack of robust empirical evaluations, with few studies utilizing experimental or longitudinal designs to assess the sustained impact of generative AI on educational outcomes. Additionally, the reviewed studies predominantly focused on English-speaking environments, limiting insights into the effectiveness of generative AI tools in multilingual and diverse cultural contexts. Ethical concerns, particularly regarding bias in AI-generated content, data privacy, and academic integrity, were raised but insufficiently addressed in the existing literature, indicating areas that require further exploration.

Looking forward, future research should prioritize the development of empirical studies that employ experimental methods and long-term assessments to better evaluate the efficacy of generative AI in education. Expanding the scope of research to encompass multilingual and cross-cultural applications of generative AI would enhance the inclusivity and global applicability of these technologies. Moreover, the creation of ethical frameworks and best practices is critical to addressing concerns over bias, data privacy, and academic integrity, which remain significant barriers to broader adoption. Additionally, integrating generative AI with adaptive learning platforms presents an exciting opportunity to create more dynamic and personalized learning experiences, offering customized support to individual students.

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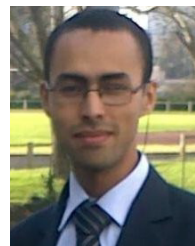
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